1.3 $T = \frac{(PSC+1)(ARR+1)}{f_{timer}}$; $T = 1s$

$f_{timer} = 8MHz = 8 \times 10^6 Hz$

$\therefore (PSC+1)(ARR+1) = 8 \times 10^6$

Solve PSC by using clock intervals of 1ms:

$T_{CLK_cnt} = \frac{(PSC+1)}{f_{timer}}$; $PSC = 1 \times 10^{-3} \cdot 8 \times 10^6 - 1$

$\therefore PSC = 7999$

Solve ARR:

$ARR = \frac{8 \times 10^6}{7999+1} - 1 = 999$

$\therefore PSC = 7999$

$ARR = 999$